

Remote Control Unit (RCU)

Hardware and Software Reference

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KJO Liquid Rocket Motor Control System

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This document describes the hardware, software architecture, operator interface, and control flow of the Remote Control Unit. The RCU is the sole operator interface to the KJO liquid rocket motor control system.

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1 Hardware Overview

1.1 Component Summary

Component	Part	Interface
Microcontroller	Adafruit Feather M0 (ATSAMD21G18A, 48 MHz)	—
Display	Adafruit 3.5" TFT V2 (HX8357D, 480×320)	SPI
Touchscreen	TSC2007 resistive controller	I ² C (0x48)
LoRa Radio	HopeRF RFM95W (Semtech SX1276), 915 MHz	SPI
Real-Time Clock	DS3231 (provides temperature also)	I ² C
SD Card	On Adafruit 3.5" TFT shield	SPI
Battery	3.7 V LiPo; voltage divider on A7	ADC A7

1.2 Pin Assignments

Signal	Pin	Notes
SCREEN_CS	11	TFT chip select
SCREEN_DC	10	TFT data/command
SD_CS	5	SD card chip select
STMPE_CS	6	Touch controller CS (also TOUCH_IRQ)
RFM95_CS	8	Radio chip select
RFM95_RST	4	Radio reset
RFM95_INT	3	Radio IRQ
BAT_SENSE	A7	Battery voltage (2:1 divider)

1.3 Hardware Block Diagram

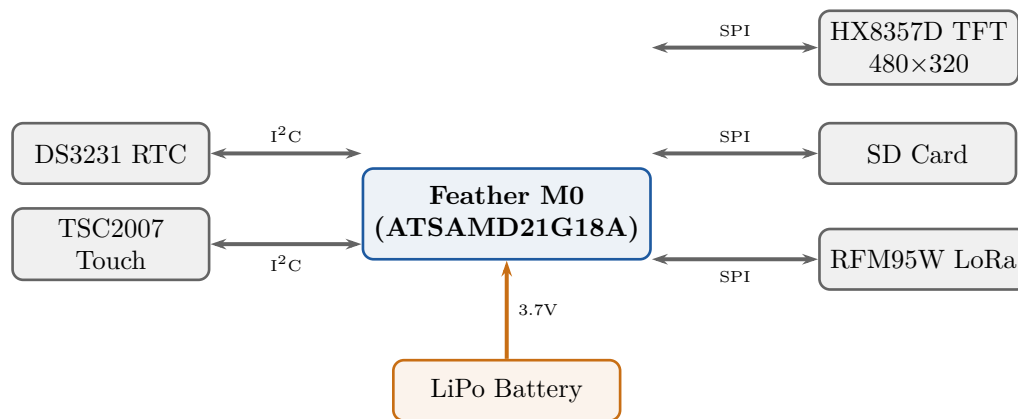


Figure 1: RCU hardware block diagram.

2 Software Architecture

2.1 Module Summary

Module	Location	Responsibility
main.cpp	local	Setup, loop, command dispatch, GUI logic
KJO_RCU_Display	shared	TFT GUI: buttons, valve symbols, status panels
KJO_Radio	shared	Radio pin constants, frequency, power
KJO_Command_Defs	shared	Command codes, packet struct definitions
KJO_System_Config	shared	Version string, platform identifier

2.2 Main Loop Control Flow

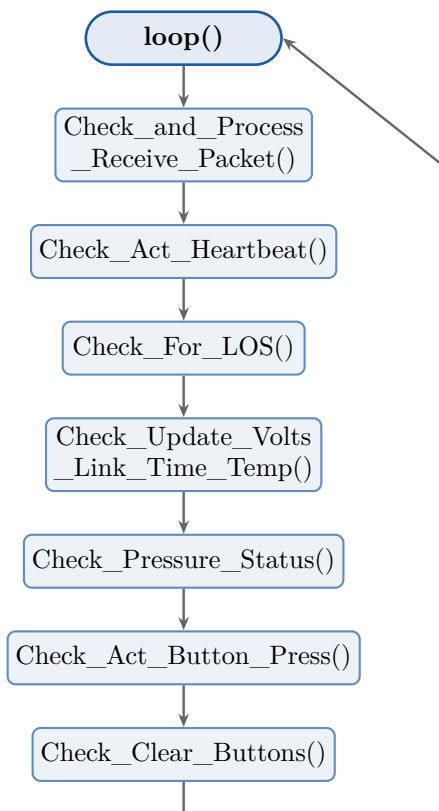


Figure 2: RCU main loop control flow.

3 Operator Interface (GUI)

3.1 Button Layout and Behaviour

Six buttons are arranged in a 2×3 grid on the left side of the screen. Each button sends one or more commands depending on its mode:

Button	Row	Col	Behaviour
Fill	1	1	Latching toggle. First press → BEGIN_FILL (button stays highlighted). Second press → STOP_FILL (button clears).
Dump	1	2	Latching toggle. First press → DUMP (button stays highlighted). Second press → STOP_DUMP (button clears).
Reset	2	1	Immediate ABORT. Sends RESET ; clears all latched button states (Fill, Dump, Enable).
Status	2	2	Sends RCU_STATUS ; auto-clears after 1 s.
Enable	3	1	Hold-to-enable LCO. Press sends ENABLE_LCO_CONTROL (param = 1, button highlighted); finger-lift sends ENABLE_LCO_CONTROL (param = 0, button clears).
Launch	3	2	Guarded by confirmation dialog. Press opens a modal CONFIRM / CANCEL overlay. CONFIRM sends START_ENGINE ; CANCEL dismisses without sending. See §4.

3.2 Valve Status Indicators

Three valve symbols occupy the right portion of the screen: Fill, N₂O (Ox), and Fuel. Each symbol uses a circle-and-bar motif: a vertical bar on a green disc indicates open; a horizontal bar on a red disc indicates closed. State is updated from the **HEARTBEAT** response packet. There is no dedicated Dump indicator — the **DUMP** command opens the Ox valve (vents pressurant through the chamber), so its state is reflected on the Ox indicator.

When the tank pressure exceeds **PRESSURE_IN_SYSTEM_THRESHOLD** (as reported in heartbeat responses), the background of the valve panel turns **red** as a visual pressure warning (see §5).

3.3 Status Panels

- **Battery voltages:** RCU voltage (measured locally on A7); EMU voltage (received in heartbeat).
- **Link status:** LINK_OK / LINK_MARGINAL / LINK_DOWN, with current RSSI value.
- **Time and temperature:** Time from DS3231; temperature converted to °F.
- **Message log:** Scrolling text panel showing last TX/RX events.

4 Engine Start Confirmation Dialog

Pressing the **Launch** button does *not* immediately send **START_ENGINE**. Instead, a modal overlay is drawn over the main screen with two large buttons:

- **CONFIRM** (green) — sends **START_ENGINE** to the EMU and dismisses the overlay.

- **CANCEL** (red) — dismisses the overlay without transmitting any command.

The overlay cannot be dismissed by touching outside of it; only **CONFIRM** or **CANCEL** can close it. This prevents an accidental single touch from initiating the ignition sequence.

5 Pressure Warning

`Check_Pressure_Status()` is called every loop iteration. It compares the most recent tank pressure value (carried in the heartbeat response packet's `r_val` field, or from the last **READ_TANK_PRESSURE** response) against `PRESSURE_IN_SYSTEM_THRESHOLD`.

When pressure exceeds the threshold:

- The valve symbol panel background is redrawn in **red**.
- A “PRESSURISED” warning label is displayed in the status area.

When pressure returns below the threshold (or a **RESET** response is received), the panel background returns to its normal colour and the warning label is cleared.

This visual warning mirrors the EMU-side audible pressure alarm (continuous buzzer on the EMU) so the operator at the RCU can immediately see that the propellant system is under pressure.

6 Radio Command Dispatch

`RCU_Execute_Command(command_ID, p1)` builds and transmits a command packet. **HEARTBEAT** uses the `Heartbeat_Command_Packet` structure (6 bytes, includes RCU voltage); all other commands use `Command_Payload` (4 bytes, includes a `short` parameter).

Responses are parsed in `Check_and_Process_Receive_Packet()`. The **HEARTBEAT** response is routed to `Process_Heartbeat_Response()`, which updates EMU voltage and all four valve symbols.

6.1 Command Reference

Command	Sent by	Notes
HEARTBEAT	Automatic (every 5 s)	RCU voltage in payload; EMU returns all valve positions
RESET	Reset button	ABORT; clears all latched button states on RCU
BEGIN_FILL	Fill button (1st press)	Starts propellant fill; EMU starts pressure recording
STOP_FILL	Fill button (2nd press)	Stops propellant fill; recording continues until ambient
DUMP	Dump button (1st press)	Opens Ox valve; vents pressurant through the chamber (no dedicated Dump valve)
STOP_DUMP	Dump button (2nd press)	Closes Ox valve
ENABLE_LCO_CONTROL	Enable button (hold / release)	param 1 = enable LCO remote start; param 0 = disable
START_ENGINE	Launch + CONFIRM	Arms 7-step ignition sequence on EMU
RCU_STATUS	Status button	Requests EMU status summary

7 SD Logging

Log files are stored in the root of the SD card as `Log_N.txt` (N is auto-incremented to avoid overwriting existing files). Each entry is time-stamped from the DS3231 and records all TX and RX events in human-readable form, e.g.:

```
10:15:03 [TX] BEGIN_FILL (1) | param: 0
10:15:03 [RX] BEGIN_FILL (1) | status: OK | r_val: 0 | RSSI: -72
```

8 Parameters & Flags

This section collects the tunable operational parameters and feature flags compiled into this firmware image, together with the source file where each is defined. It is meant as a single reference point when tracing or changing a timing, threshold, or calibration value. Pin/channel wiring assignments, LoRa command opcodes, and packet byte layouts are documented separately (the Hardware Overview pin map and the Command Reference sections above) and are not repeated here. Parameters marked (*shared*) are defined in `KJO_Shared_Libraries` and must agree with the same constant used on the other board(s) that consume it.

Radio Communication

Parameter	Value	Description	Source File
RCS_CENTER_FREQUENCY	917.117 MHz	LoRa carrier frequency; must match on both RCU and EMU radios to communicate.	<code>KJO_Radio.h</code> (shared)
TRANSMIT_POWER	5 dBm	RFM95 transmit power (PA_BOOST pin), valid range 5–23 dBm.	<code>KJO_Radio.h</code> (shared)

LINK_MARGINAL_THRESHOLD	−90 dBm	RSSI floor below which the link status indicator shows MARGINAL (yellow) instead of OK (green).	KJO_Radio.h (shared)
MAX_PAYLOAD_LENGTH	64 bytes	Size of the LoRa TX/RX payload buffer; sized with headroom above the largest packet (<code>Extended_Heartbeat_Response_Packet</code> , 39 bytes).	KJO_Command_Defs.h (shared)
heartbeat_interval	5000 ms	Interval at which a HEARTBEAT command is sent to the EMU when no other traffic has occurred, to keep the radio link alive.	main.cpp
LOS_timeout_interval	30000 ms	Time since last received packet after which the link is declared LOST (LINK_DOWN) and RSSI/status reset.	main.cpp
FILL_CMD_RETRY_DELAY_MS	250 ms	Delay before firing the single automatic <code>BEGIN_FILL/STOP_FILL</code> retry after EMU responds <code>RELAY_BUSY</code> .	main.cpp

GUI / Display Timing

Parameter	Value	Description	Source File
volt_RSSI_update_interval	1000 ms	Refresh interval for the battery voltage, link status, time, and temperature display fields.	main.cpp
show_button_duration	1000 ms	How long a momentary button stays highlighted (pressed color) before reverting, for non-latching buttons.	main.cpp

Fill Progress Polling

Parameter	Value	Description	Source File
FILL_PROGRESS_STATUS_INTERVAL	1500 ms	Poll interval for <code>QUERY_FILL_PROGRESS</code> while a fill session is active, driving the Fill button color and popup progress bar.	main.cpp
PRESSURE_STATUS_INTERVAL	2000 ms	Poll interval for <code>READ_TANK_PRESSURE</code> while a fill is active or the pressure warning is active.	main.cpp

Pressure Monitoring

Parameter	Value	Description	Source File
PRESSURE_IN_SYSTEM_THRESHOLD	10 PSI	Tank pressure above which the RCU displays the “system pressurized” warning (valve panel turns red).	KJO_RCU_Display.h (shared)

SD Logging

Parameter	Value	Description	Source File
LOG_FILE_NAME_BASE	"Log_"	Base filename prefix for SD-card log files; combined with an auto-incrementing index (<code>Log_0.txt</code> , <code>Log_1.txt</code> , ...) to find the next unused file at boot.	main.cpp

Battery Voltage Calibration (RCU Local ADC)

Parameter	Value	Description	Source File
RCU_Batt_Info.AD_res	12 bits	ADC resolution used when sampling the RCU's own battery voltage divider on pin A7.	main.cpp
RCU_Batt_Info.AD_samples	4	Number of ADC samples averaged per RCU battery voltage reading (noise reduction).	main.cpp

9 Version History

Version	Date	Notes
0.1	March 2026	Initial release.
0.2	March 2026	Button table updated: FILL and DUMP are latching toggles; Reset is immediate ABORT and clears all latched states; Enable is hold-to-enable LCO (lift = disable); Launch guarded by confirmation dialog. Main loop diagram updated to include <code>Check_Pressure_Status()</code> . New §4 Engine Start Confirmation Dialog. New §5 Pressure Warning (valve panel background turns red).
0.3	March 2026	SD log files moved to SD card root (<code>Log_N.txt</code>); subdirectory path constant (<code>LOG_FILE_FOLDER</code>) removed from <code>main.cpp</code> .
0.4	July 2026	Dump valve physically removed from the vehicle. <code>DUMP/STOP_DUMP</code> retained (still a latching toggle) but now open/close the Ox valve instead of a dedicated Dump valve; the separate Dump valve status indicator was removed and its state folded into the Ox indicator.
0.5	September 2026	Add new §8 Parameters & Flags reference section: tunable timing, threshold, and calibration constants across the radio, GUI, fill-progress polling, and pressure-monitoring subsystems, with source-file attribution for each.